

Antituberculosis Drugs

Tuberculosis

- Two species of Mycobacterium cause tuberculosis:
 - ✓ M. tuberculosis
 - ✓ M. bovis
- M. tuberculosis is transmitted by inhalation of infective droplets **coughed** or **sneezed** into the air by a patient with tuberculosis.
- M. bovis is transmitted by **milk** from diseased cows and first produces intestinal or tonsillar lesions.

Tuberculosis

- Mycobacteria grow slowly and may be **dormant** in the host for long periods
- Characteristics of mycobacteria that make the diseases chronic & necessitate prolonged treatment
- Many antibacterials do not penetrate mycobacterial cell walls, and a portion reside inside macrophages
- Mycobacteria are alert in developing resistance to single chemotherapeutic agents
- Consequence = effective therapy requires a prolonged course of multiple drugs
 - ✓ Patient compliance, drug toxicity, drug interactions, concurrence of other diseases

Antituberculosis Drugs

❖ **First-line** drugs are **superior in efficacy** and possess an **acceptable degree of toxicity**.

- Include isoniazid, rifampin, pyrazinamide, ethambutol, and streptomycin.
- Most patients with tuberculosis can be treated successfully with these drugs.

❖ **Second-line** drugs are **more toxic** and **less effective**

- Indicated only when the M. tuberculosis organisms are resistant to the first-line agents.
- Include cycloserine, ethionamide, aminosalicylic acid, rifabutin, quinolones, capreomycin, viomycin, and thiacetazone.

First-line drugs for tuberculosis

Isoniazid

- Antibacterial Activity
- The most active drug for the treatment of tuberculosis caused by susceptible strains
- Is **bacteriostatic** for "resting" bacilli, but is **bactericidal** for rapidly dividing microorganisms
- Remarkably selective for mycobacteria
- Penetrates cells with ease and is just as effective against bacilli growing within cells as it is against those growing in culture media (Effective intracellular activity)

Isoniazid

- Mechanism of Action
- Mycobacterial catalase-peroxidase (katG encoded) converts isoniazid into an active metabolite
- A primary action of isoniazid is to inhibit the biosynthesis of mycolic acids (MA)
- Resistance can be mutations in at least five different genes:
- Catalase-peroxidase-activated isoniazid binds enoyl-ACP reductase of fatty acid synthase II, which converts unsaturated fatty acids to saturated fatty acids in the mycolic acid biosynthetic pathway.
- It also inhibits mycobacterial catalase-peroxidase, which may increase the likelihood of damage to the mycobacteria from reactive oxygen species and H₂O₂.

Isoniazid

- Pharmacokinetics
- Water soluble, well absorbed from oral & parenteral sites
- Does not bind to serum proteins, diffuses readily into all body fluids and cells
- Concentrations in CNS & CSF are similar to those in plasma during meningeal inflammation
- INH is acetylated to acetyl INH by N-acetyltransferase
- Genetic polymorphism in the rate of acetylation.
- Slow or rapid acetylation is rarely important clinically, although slow acetylators tend to develop **peripheral neuropathy** more readily.

Isoniazid

- **Clinical Uses**
- Isoniazid is the safest and most active mycobactericidal
- Considered the primary drug for use in all therapeutic and prophylactic regimens for susceptible tuberculosis infections
- As a single agent for treatment of latent tuberculosis, which is usually determined by a positive tuberculin skin test.
- TB prophylaxis in HIV/AIDS patients

Isoniazid

❖ Adverse Reactions

- Allergic Reactions

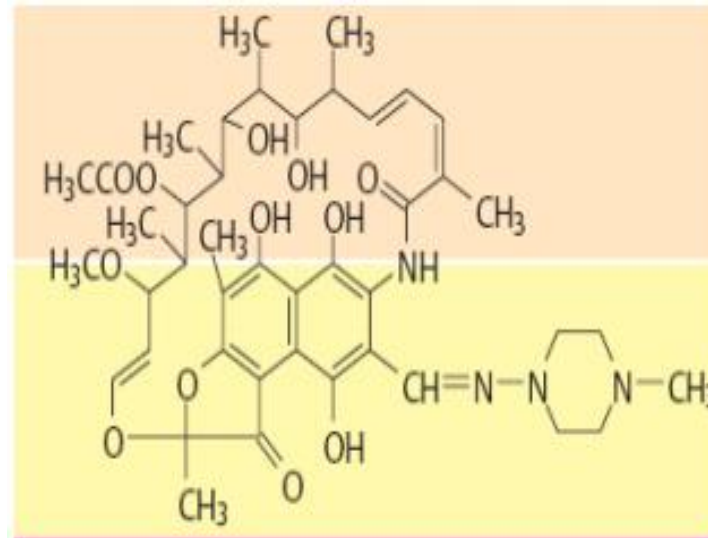
- ✓ Fever, skin rashes, Drug-induced systemic lupus erythematosus

- Direct Toxicity

- ✓ **Hepatitis** (greater risk in alcoholics & pregnancy and is a contraindication to further use of the drug).
- ✓ **Peripheral neuropathy** due to a relative pyridoxine deficiency (more likely in slow acetylators, malnutrition, alcoholism, diabetes, AIDS)
 - Minimized by administration of pyridoxine
- ✓ **CNS toxicity** (memory loss, psychosis, and seizures)
 - May respond to pyridoxine

Rifampin and other rifamycins

- The rifamycins (rifampin, rifabutin, rifapentine) are a group of structurally similar, complex macrocyclic antibiotics produced by *Ammycolatopsis mediterranei*
- Rifampin is a semisynthetic derivative of rifamycin B.
- Rifampin is soluble in organic solvents and in water at acidic pH.



Rifampin

- **Antibacterial Activity**

- Rifampin inhibits the growth of *M. tuberculosis* in vitro.
- It also inhibits the growth of non-tuberculous mycobacteria
- Rifampin inhibits the growth of most Gm(+) bacteria as well as many Gm(-) microorganisms such as *E. coli*, *Pseudomonas*, *Proteus*, and *Klebsiella*.

- **Mechanism of Action**

- Rifampin binds DNA-dependent **RNA polymerase** and forms stable drug-enzyme complex, leading to suppression of initiation of chain formation in RNA synthesis.

Rifampin

- Nuclear eukaryotic RNA polymerases are **unaffected** but High concentrations can inhibit RNA synthesis in mammalian
- Bactericidal for both intracellular and extracellular microorganisms.
- **Clinical Uses**
 - A first-line antitubercular drug used in the treatment of all forms of pulmonary and extrapulmonary TB
 - Alternative to INH in the treatment of latent TB infection
 - In combination with other **antileprosy** agent for the treatment of leprosy
 - Also used in **MRSA** infections

Rifampin

- **Pharmacokinetics**

- Well absorbed after oral administration and excreted mainly through the liver into bile.
- Undergoes enterohepatic recirculation
- Intestinal reabsorption is reduced by food
- Small amount excreted in urine
- Distributed widely in body fluids and tissues
- Rifampin is relatively highly protein-bound,
- Adequate CSF concentrations are achieved only in the presence of meningeal inflammation

Rifampin

- **Adverse Reactions**

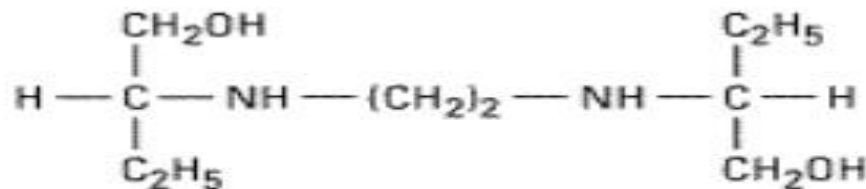
- ✓ Imparts **orange color** to urine, sweat, tears
- ✓ Rashes, thrombocytopenia, and nephritis, cholestatic jaundice and occasionally hepatitis
- ✓ It is a strong **inducer** of most cytochrome P450 isoforms
- ✓ Increases the elimination of numerous drugs including methadone, anticoagulants, some anticonvulsants, protease inhibitors, and contraceptives

Ethambutol

- Ethambutol is a synthetic, water-soluble, heat-stable compound

❖ Antibacterial Activity

- Nearly all strains of *M. tuberculosis* and *M. kansasii*
- Ethambutol has no effect on other bacteria.
- It suppresses the growth of most INH- and streptomycin-resistant tubercle bacilli.
- Resistance to ethambutol develops very slowly in vitro.



Ethambutol

Ethambutol

- **Mechanism of Action and Resistance**
- Inhibits arabinosyl transferases involved in cell wall biosynthesis.
- The enzyme is involved in the polymerization reaction of arabinoglycan, an essential component of the mycobacterial cell wall.
- Bacterial resistance to the drug develops via single amino acid mutations in the embA gene when ethambutol is given in the absence of other effective agents

Ethambutol

- **Pharmacokinetics**

- ✓ Well absorbed from the GIT
- ✓ About 20% of the drug is excreted in feces and 50% in urine in unchanged form (dose adjustment in renal impairment)
- ✓ Is widely distributed in all body fluids
 - It crosses the BBB only if the meninges are inflamed.

Ethambutol

- **Clinical Uses**
- Has replaced aminosalicylic acid as a first-line antitubercular drug.
- Commonly included as a 4th drug, along with INH, pyrazinamide, and rifampin, in patients infected with MDR strains.
- Also used in combination in the treatment of M. avium-intracellulare infection in AIDS patients.

Ethambutol

- **Adverse Reactions**
- Hypersensitivity , **loss of visual acuity** and red-green color blindness (most serious side-effects)
- Ethambutol is relatively contraindicated in children too young to permit assessment of **visual acuity** and **red-green color** discrimination.

Streptomycin

- An aminoglycoside antibiotic and was the first drug to reduce tuberculosis mortality.
- It is bactericidal against *M. Tuberculosis* in vitro but is inactive against intracellular organisms.
- Most *M. tuberculosis* strains and non-tuberculosis species, such as *M. kansasii* and *M. Avium intracellulare*, are sensitive.

Streptomycin

❖ Antibacterial Activity

- Bactericidal in vitro.
- Suppressive activity in vivo
- Most *M. tuberculosis* strains and non-tuberculosis species, such as *M. kansasii* and *M. aviumintracellulare*, are sensitive
- It penetrates into cells poorly, and is active mainly against extracellular tubercle bacilli

Streptomycin

- **Mechanism of action**
- Prokaryotic ribosomal protein synthesis inhibitor
- Resistance is related to a point mutation that involves the gene (rpsl or rrs) that encodes for ribosomal proteins and binding sites
- About 80% of strains that are resistant to INH and rifampin are also resistant to streptomycin.

Streptomycin

- **Clinical Use in TB**
- As a fourth drug in combination with INH, rifampin, and pyrazinamide in patients at high risk for drug resistance
- Employed when an injectable drug is needed or desirable, (severe, possibly life-threatening forms of TB, and in treatment of infections resistant to other drugs)
- Other drugs are always given simultaneously to prevent emergence of resistance

Pyrazinamide

- Synthetic pyrazine analog of nicotinamide
- It is stable, slightly soluble in water, and quite inexpensive.
- At neutral pH, it is inactive in vitro, but at pH 5.5 it inhibits tubercle bacilli
- Drug is taken up by macrophages and exerts its activity against intracellular organisms residing within acidic environment.



Pyrazinamide (PZA)

Pyrazinamide

- **Mechanism of Action and Resistance**

- Exact MOA unknown, although its target appears to be the mycobacterial fatty acid synthetase involved in mycolic acid biosynthesis.
- It is a prodrug activated to the active pyrazinoic acid by the mycobacterial enzyme pyrazinamidase
- Mutation in the gene (pncA) that encodes pyrazinamidase is responsible for drug resistance
- Resistance develops **fairly readily**, but there is no cross-resistance with INH or other antimycobacterial drugs.

Pyrazinamide

- **Pharmacokinetics**

- Well absorbed from the GIT and widely distributed in body tissues, including inflamed meninges

- **Clinical uses**

- An important front-line drug used in conjunction with INH and rifampin in short-course (ie, 6-month) regimens as a "sterilizing" agent active against residual intracellular organisms that may cause relapse.

Adverse effects

- **Hepatotoxicity** (in 1–5%), nausea, vomiting, drug fever, and hyperuricemia (occurs uniformly and may provoke acute gouty arthritis)

Alternative Second-Line Drugs for Tuberculosis

Quinolones

- The fluoroquinolones are highly active against *M. tb* as well as non-tuberculous mycobacteria
- important components of treatment regimens of MDR TB
 - gatifloxacin and moxifloxacin are the most active
- Cross-resistance within the entire class is common
- To prevent development of resistance, the most active fluoroquinolones should be used, and only in combination with other antimycobacterial agents

Ethionamide

- A derivative of isonicotinic acid and is chemically related to isoniazid.
- A secondary agent used in combination when primary agents are ineffective or contraindicated
- It is a bacteriostatic antituberculosis agent.
- **Mechanism of Action**
- Inhibits the activity of the inhA gene product, the enoyl-ACP reductase of fatty acid synthase II
- Inhibition of mycolic acid biosynthesis and consequent impairment of cell-wall synthesis.

Ethionamide

- **Pharmacokinetics**

- Well absorbed following oral administration
- Rapidly and widely distributed to all body tissues and fluids, including the cerebrospinal fluid.
- Cleared by hepatic metabolism; it inhibits the acetylation of isoniazid in vitro.
- Less than 1% of the drug is eliminated in the urine unchanged.

Ethionamide

- **Therapeutic Uses**

- A secondary agent, used concurrently with other drugs only when therapy with primary agents is ineffective or contraindicated.

- **Untoward Effects**

- Anorexia, nausea and vomiting, gastric irritation, and a variety of neurologic symptoms.
- Severe postural hypotension, mental depression, drowsiness, convulsions, peripheral neuropathy.
- Pyridoxine (vitamin B6) relieves the neurologic symptoms
- Metallic taste

P-Aminosalicylic acid

- Folate synthesis antagonist that is active almost exclusively against M. tb.
- It is structurally similar to p-aminobenzoic acid (PABA) and to the sulfonamides
- **Mechanism of Action**
- It is a folate synthesis antagonist that interferes with the incorporation of PABA into folic acid. PAS is **bacteriostatic**
- **Therapeutic Uses**
- A second-line agent

Untoward Effects

- GI problems(anorexia, nausea, epigastric pain, abdominal distress, and diarrhea)
- Patients with peptic ulcers tolerate the drug poorly
- Hypersensitivity reactions
- Hematological abnormalities (leukopenia, agranulocytosis, eosinophilia, lymphocytosis, thrombocytopenia).

Cycloserine

- Cycloserine is a broad-spectrum antibiotic produced by *Streptomyces orchidaceus*.
- It is structural analogue of D-alanine
- **Antibacterial Activity**
- **No cross-resistance** between cycloserine and other tuberculostatic agents.
- D-alanine blocks the antibacterial activity of cycloserine
- Inhibitory to *M. tb* and active against *E. coli*, *S. aureus*, and *Enterococcus*, *Nocardia*, and *Chlamydia* spp.

- **Mechanism of Action**

- Cycloserine and D-alanine are structural analogs; thus, cycloserine inhibits reactions in which D-alanine is involved in **bacterial cell-wall synthesis**

- **Pharmacokinetics**

- Rapidly absorbed When given orally (70% to 90%)
- Distributed throughout body fluids and tissues
- CSF concentrations are approximately the same as those in plasma
- About 65% of a parenteral dose is excreted unchanged in the urine in the first 72 hours.

- **Therapeutic Uses**

- In the treatment of MDR tuberculosis and is useful in renal tuberculosis, since most of the drug is excreted unchanged in the urine

- **Untoward Effects**

- CNS (headache, tremor, vertigo, confusion, nervousness, irritability, psychotic states with suicidal tendencies, paranoid reactions, catatonic and depressed reactions, visual disturbances, seizures)
- Contraindicated in individuals with a history of **epilepsy** and should be used with caution in individuals with a history of **depression**, as suicide is a risk.

Other drugs

- **Kanamycin & Amikacin**

- ✓ Aminoglycoside antibiotics

- Amikacin is indicated for treatment of tb suspected or known to be caused by streptomycin-resistant or MDR strains

- **Capreomycin**

- A polypeptide antibiotic derived from *Streptomyces capreolus*.
- It is **bacteriostatic** against most strains of *M. tuberculosis*, including the MDR strain.
- Used as a second-line agent in combination with other drugs. Particularly useful in multidrug regimens for the treatment of drug resistant tb (esp with streptomycin resistance)
- Capreomycin is associated with **ototoxicity** and **nephrotoxicity**

Chemotherapy for lepros

- Leprosy (Hansen's disease) is rare in the United States, but a small number of cases, both imported and domestically acquired, are reported each year.
- Approximately 70 percent of all cases in the world are located in India.
- Bacilli from skin lesions or nasal discharges of infected patients enter susceptible individuals via abraded skin or the respiratory tract.
- The World Health Organization recommends the triple-drug regimen of **dapsone, clofazimine, and rifampin** for 6 to 24 months.

A. Dapsone

- Dapsone is structurally related to the sulfonamides
 - similarly inhibits **folate** synthesis via dihydropteroate synthetase inhibition.
- It is **bacteriostatic** for *Mycobacterium leprae*, but resistant strains are encountered.
- Dapsone is also employed in the treatment of pneumonia caused by *Pneumocystis jirovecii* in patients infected with HIV.
- The drug is well absorbed from the gastrointestinal tract
- is distributed throughout the body, with high levels concentrated in the skin.

Dapsone

- The parent drug enters the enterohepatic circulation and undergoes hepatic acetylation.
- Both parent drug and metabolites are eliminated through the urine.
- Adverse reactions include hemolysis, methemoglobinemia, **peripheral neuropathy**, and the possibility of developing erythema nodosum leprosum (a serious and severe skin complication of leprosy).
- The latter is treated with corticosteroids or thalidomide.

B. Clofazimine

- Clofazimine is a phenazine dye that binds to DNA and prevents it from serving as a template for future DNA replication.
- Its redox properties may lead to the generation of cytotoxic oxygen



Figure 34.13
Leprosy patient. A. Before therapy.
B. After 6 months of multidrug
therapy.